P84 推导

stoic

cynic

~~非相型~~ 非相变量型

$$\begin{cases} \dot{z} = Az + Bu \\ y = Cz \end{cases}$$

Control
可控

$$Q_{cz} = [B;\ AB;\ A^2B \cdots A^nB]$$

引入矩阵 P

$$z = Px \leftarrow \text{相变量型}$$

$$\begin{cases} P\dot{x} = AP\dot{x} + Bu \\ y = CPx \end{cases}$$

$$\begin{cases} \dot{x} = P^{-1}AP\cdot x + P^{-1}B\cdot u \\ y = CPx \end{cases}$$

可控 $Q_{cx} = [P^{-1}B;\ (P^{-1}AP)(P^{-1}B);\ (P^{-1}AP)^2(P^{-1}B) \cdots (P^{-1}AP)^{n-1}(P^{-1}B)]$

~~Q_{cz}~~ . Q_{cx} $\Rightarrow$ (n – ~~a 阶~~ 的 B (3×1)) $n=3$

$$[B,\ AB,\ A^2B]\begin{bmatrix} P^{-1}B \\ \cancel{P^{-1}AB} \\ P^{-1}A^2B \end{bmatrix}$$

不是转置

$$(P^{-1}A\cancel{P})(\cancel{P^{-1}}A\cancel{P})(\cancel{P^{-1}}B)$$
$$= P^{-1}A^2B$$

~~$[B,\ AB,\ A^2B]$~~

$$Q_{cx} = [P^{-1}B,\ P^{-1}AB,\ P^{-1}A^2B]$$
$$= P^{-1}[B,\ AB,\ A^2B]$$
$$= P^{-1}\ \cancel{Q_{cx}}\ Q_{cz}$$

~~$P = Q_{cz}\ Q_{cx}$ $Q_{cx}^{-1}\ Q_{cz}$~~

$$P\,Q_{cx} = Q_{cz}$$

~~$Q\,P$~~

$$P\cdot Q_{cx}\cdot Q_{cx}^{-1} = Q_{cz}\,Q_{cx}^{-1} \Rightarrow P = Q_{cz}\cdot Q_{cx}^{-1}$$

非相变量型　　相变量型
通过P矩阵将 z空间转到 x空间后.

$$\begin{cases} \dot{x} = P^{-1}APx + P^{-1}Bu \\ y = CPx \end{cases}$$

在相变量型 x 中加入反馈 -K

$$u \to r - Kx$$

$$\begin{cases} \dot{x} = P^{-1}APx + P^{-1}B(r - Kx) \\ \quad = (P^{-1}AP - P^{-1}BK_x)x + P^{-1}Br \\ y = CPx \end{cases}$$

再从~~x~~转回z. 通过 $z = Px$

即 $x = P^{-1}z$

~~$P^{-1}\dot{z} = P^{-1}AP \cdot P^{-1}z +$~~

$$\begin{cases} P^{-1}\dot{z} = (P^{-1}AP - P^{-1}BK_x) \cdot P^{-1}z + P^{-1}Br \\ y = CP \cdot P^{-1}z = Cz \end{cases}$$

$$\dot{z} = P(P^{-1}AP - P^{-1}BK_x) \cdot P^{-1}z + Br$$

$$\begin{cases} \quad = (AP - BK_x)P^{-1}z + Br \\ \quad = (A - BK_xP^{-1})z + Br \\ y = Cz \end{cases}$$

⑩

$$\begin{cases} \dot{z} = (A - BK_x P^{-1}) z + Br \\ y = Cz \end{cases}$$

对此如果是修改 K_z

$$\begin{cases} \dot{z} = (A - BK_z) z + Br \\ y = Cz \end{cases}$$

$$\cancel{B} K_x P^{-1} = K_z$$